Pipelines and analytics

Microsoft Fabric

C4 Ingestion and orchestration

C4-L1 Choosing how data gets into Fabric

Mirroring for curated databases, Copy job for bronze, Eventstream for events, Spark or Dataflow Gen2 for transformation, and pipelines to orchestrate.

Microsoft's starting rule by layer	Gold from existing systems: Mirroring. Bronze ingestion: Copy job. Low-latency events: Eventstream.
What a pipeline is for	Orchestration: ordering, branching, loops, retries and alerts across activities.
Spark or Dataflow Gen2?	Spark for code-first, large-volume engineering. Dataflow Gen2 for low-code Power Query authors.
Copy activity vs Copy job for incremental loads	Copy activity needs hand-built state. Copy job tracks the last successful run itself.

C4-L2 Incremental copy with a watermark, or with CDC

Watermarks copy rows above the last value seen: no deletes, no late rows. Use CDC or a ROWVERSION column, and Merge for one row per key.

Copy job incremental copy	First run copies everything. Later runs copy new or changed data. State is tracked by the job.
Watermark blind spots	Cannot see deletes. Misses late rows at or below the stored watermark.
When to use CDC	When deletes must replicate, or changes arrive out of order.
Copy job update methods	Append (default), Merge, Overwrite, SCD Type 2. Merge for one row per key.

C4-L3 Pipelines that fail fast, retry sensibly and alert

Set timeouts near the real run time, retry only transient failures, chain on Upon Success and alert on Upon Failure.

Activity timeout default	12 hours, maximum seven days, format D.HH:MM:SS. Set it near the real run time.
Activity retries	Only when enabled. Count defaults to 1, interval 30 seconds. Retry transient failures only.
Dependency branches	Upon Success, Upon Failure, Upon Completion (either way), Upon Skip.
@{...} vs @...	@{...} inside text always yields a string. A bare @ expression keeps the value's type.

C5 Spark notebooks in Fabric

C5-L1 From Databricks notebooks to Fabric notebooks

Replace dbutils with notebookutils, secrets with Key Vault, inline installs with environments, and pass real timeouts to child notebooks.

Fabric replacement for dbutils	notebookutils (formerly mssparkutils).
Secrets in Fabric notebooks	notebookutils.credentials.getSecret(vault_url, name), from Azure Key Vault.
notebookutils.notebook.run default timeout	90 seconds. Pass timeout_seconds.
%pip in pipeline runs	Turned off by default. Use an environment.

C5-L2 Environments, resource profiles and V-Order

Pin libraries in a Full mode environment, keep the writeHeavy default for ingestion, and turn V-Order on for read-heavy gold tables.

Environment publishing modes	Quick: installs at session start (adds minutes). Full: reproducible snapshot, recommended for pipelines.
V-Order in new workspaces	Off by default (writeHeavy profile). Write cost around 15%, faster reads.
Turn V-Order on for one table	TBLPROPERTIES("delta.parquet.vorder.enabled" = "true").
spark.sql.parquet.vorder.enable	Removed in Runtime 1.3 and later. Use spark.sql.parquet.vorder.default or table and writer options.

C5-L3 Delta tables shared between Databricks and Fabric

Open format, but not every feature for every reader. Design shared gold tables for the weakest reader that needs them.

Delta interoperability rule	Every reader must support every feature the writer put in the table.
TIMESTAMP_NTZ in Fabric	Spark yes. SQL analytics endpoint and Direct Lake no.
Runtime 2.0 default table protocol	Reader 3, writer 7, deletion vectors on. Python notebooks cannot read deletion vectors.
Check a table's features	DESCRIBE DETAIL: minReaderVersion, minWriterVersion, table features.

C6 Semantic models and Direct Lake

C6-L1 Star schemas for semantic models

Facts with a stated grain, dimensions with unique keys, one-to-many single-direction relationships, and derived columns computed upstream.

Star schema parts	Fact tables (keys and measures, with a grain) and dimension tables (unique key, descriptive columns).
Relationship guidance	One-to-many, single direction. Minimise bi-directional relationships.
Calculated columns and Direct Lake	Not supported on SQL endpoints, preview on OneLake. Compute derived columns in gold.
Grain	Set by the fact table's dimension keys. State it in the table's docstring and comment.

C6-L2 DAX measures that state their filter context

Measures, not calculated columns. DIVIDE for every ratio, variables for clarity, and CALCULATE with the narrowest filter change.

```
Weighted aware % =
VAR aware_w = SUMX ( fact_awareness, fact_awareness[aware] * fact_awareness[weight] )
VAR base_w  = SUM ( fact_awareness[weight] )
RETURN DIVIDE ( aware_w, base_w )
```

Measure vs calculated column	Measures compute at query time in the visual's filter context. Calculated columns are stored at refresh.
DIVIDE in DAX	Returns BLANK (or an alternate result) when the denominator is zero or BLANK.
VAR ... RETURN	Evaluates once, names parts, improves performance, readability and debugging.
Share of total across brands	CALCULATE([measure], ALL(dim_brand)): remove only the brand filter.

C6-L3 Direct Lake, framing and the fallback you did not plan

Direct Lake stays fast when tables are Delta (not views), compact, within the SKU's guardrails, and fallback is made visible in development.

Direct Lake refresh	Framing: records Delta table versions in seconds. No data copy.
Direct Lake on OneLake vs on SQL	OneLake: never falls back. SQL: falls back to DirectQuery for views and SQL-based security.
Direct Lake behavior values	Automatic (default, silent fallback), DirectLakeOnly (error), DirectQueryOnly (always DirectQuery).
F64 per-table guardrails	5,000 Parquet files, 1.5 billion rows, 25 GB max memory.

C7 Real-Time Intelligence

C7-L1 Eventstream, events in, routed out

Land the raw stream once, derive filtered streams for consumers, and route each to the destination that suits it.

What an eventstream does	Ingests real-time events, transforms them without code, and routes them to destinations.
Eventstream destinations	Eventhouse (direct or processed ingestion), Lakehouse (Delta tables), Activator, custom endpoint.
Permission for eventhouse, lakehouse or Activator destinations	Contributor or higher on the destination workspace.
Protect replay	Land the raw stream first. Derive filtered streams from it.

C7-L2 Eventhouse and KQL for time-series questions

Filter on time first, use top for "latest", make-series for continuous charts, and set retention and caching per table on purpose.

```
telemetry_raw
| where event_time > ago(1h)
| where event_type == "reading"
| summarize avg_rpm = avg(rpm), n = count() by vehicle_id, bin(event_time, 5m)
```

take vs top in KQL	take: arbitrary rows, not repeatable. top N by col desc: ordered, repeatable.
Gaps in time series	summarize omits empty bins. make-series ... default = 0 fills them.
KQL retention policy defaults	Default 3,650 days, maximum 36,500, minimum 1 day, or unlimited.
Caching policy	How much recent data stays in the hot SSD cache: faster queries, higher cost.

C7-L3 Activator, rules that act on events

Key each rule by the object it describes. Fire on change or sustained state, not on every event. Add a heartbeat for silence.

Events and objects in Activator	Events are grouped into objects by a shared ID. Rules run and keep state per object.
Increases above vs Is greater than	Increases above fires on the crossing. Is greater than fires on every matching event.
Detect a silent sensor	No presence of data (heartbeat) fires when no events arrive within a set time.
Activator limits to design around	10,000 events per second per rule. 30 emails or Teams messages per rule per recipient per hour.